CS 3410 Software Engineering

Assignment 2 — Architecture and Quality Plan for the Campus Library Reservation System

Semester: Fall 2026 **|** Weight: 25% of final grade **|** Individual assignment **|** Due: Friday, Week 9, 23:59 (local time)

Instructor: Course staff **|** Questions: course discussion board only (no email submissions)

1. Overview

The university library is replacing its ageing study-room reservation tool with a new Campus Library Reservation System (CLRS). You have been hired as the lead engineer for the first release. Your task in this assignment is NOT to implement the full system; it is to produce an engineering design document that selects an architectural style, justifies it against the quality requirements, and defines the quality plan (testing, risks, delivery) that the implementation team will follow in Assignment 3.

This assignment assesses your ability to reason about trade-offs. There is no single correct architecture; marks are awarded for the quality of the justification, the consistency between the chosen architecture and the quality plan, and the clarity of the document.

2. Learning outcomes assessed

- LO2 — Compare architectural styles and select one appropriate to stated functional and quality requirements.
- LO3 — Derive a testing strategy (unit, integration, contract, end-to-end) from architectural decisions.
- LO4 — Identify technical and project risks and propose mitigations with measurable triggers.
- LO6 — Communicate engineering decisions in a structured design document.

3. System description

CLRS lets students and staff reserve study rooms and equipment across three campus libraries. It must integrate with the existing university identity provider (SAML), the facilities calendar (iCal feeds), and a legacy room-inventory database (PostgreSQL, read-only). Library staff manage rooms, opening hours and block bookings; students search availability, reserve, modify and cancel; kiosk displays at each room show the current and next reservation.

3.1 Functional requirements (excerpt)

- FR-1 A user can search available rooms by campus, capacity, equipment and time window.
- FR-2 A user can create, modify and cancel a reservation; reservations are limited to 3 hours per day per user.
- FR-3 Staff can create recurring block bookings that override student reservations with 48 hours' notice.
- FR-4 Kiosk displays refresh within 10 seconds of any reservation change for their room.
- FR-5 Users receive reminders (email/push) 15 minutes before a reservation starts.
- FR-6 Staff can export monthly utilisation reports per campus.

3.2 Quality requirements

ID	Attribute	Requirement
QR-1	Availability	99.5% monthly during opening hours; kiosks degrade gracefully to last-known schedule when offline.

QR-2	Performance	Availability search p95 under 400 ms with 2,000 concurrent users at semester start.
QR-3	Security	All user actions authenticated via SAML; staff actions audited; no PII in logs.
QR-4	Modifiability	A new library (campus) must be onboardable with configuration only, no code change.
QR-5	Testability	The reservation rules must be testable without a database or network.
QR-6	Operability	Deployable by a two-person team; rollback within 10 minutes.

4. Your task

Produce a design document (maximum 3,000 words, excluding diagrams and appendices) that addresses the items below. Use the section structure given in 4.3; markers follow it when grading.

4.1 Architectural style selection

Evaluate the following candidate styles for CLRS and select ONE as your primary architectural style:

- Method A — Layered monolith (presentation / application / domain / persistence layers in a single deployable).
- Method B — Microservices (separate reservation, inventory, notification and reporting services with an API gateway).
- Method C — Hexagonal architecture (ports and adapters around a framework-independent domain core).
- Method D — Event-driven architecture with CQRS (commands, an event log and read-model projections for search and kiosks).

Your justification must reference at least four of the quality requirements in 3.2 and explain what you give up by not choosing the alternatives. A choice without an explicit trade-off discussion receives at most half marks for this section.

4.2 Quality plan

- Testing strategy: which kinds of tests exist, what each kind covers, and how the architecture makes them possible (e.g., how QR-5 is satisfied).
- Delivery: branching, continuous integration gates, deployment and rollback approach appropriate for QR-6.
- Risks: at least five technical or project risks, each with likelihood, impact, a measurable trigger and a mitigation.
- Observability: what you log, measure and alert on to know that QR-1 and QR-2 are met in production.

4.3 Required document structure

- 1. Context and scope (max. 300 words)
- 2. Architectural decision (style, rationale, rejected alternatives)
- 3. Structure (component or container diagram plus a short walkthrough of FR-1 and FR-4)
- 4. Quality plan (testing, delivery, observability)
- 5. Risks and mitigations
- 6. Open questions for the product owner

5. Deliverables and submission

- One PDF named <studentid>-a2.pdf uploaded to the course LMS before the deadline.
- Diagrams may be hand-drawn and photographed if legible; use C4 or UML notation where possible.
- Late submissions lose 10% per calendar day; extensions only via the standard special-consideration process.

6. Marking rubric

Criterion	Weight	Excellent (HD)	Adequate (P)
Architectural decision and trade-offs	35%	Decision clearly tied to QR-1..QR-6; rejected options analysed honestly	Decision stated; thin or generic rationale
Structure and walkthroughs	15%	Diagram and walkthroughs consistent with the chosen style	Diagram present but inconsistent with prose
Quality plan	25%	Tests, delivery and observability follow from the architecture	Generic checklist not linked to the design
Risks	15%	Specific, measurable triggers and realistic mitigations	Vague risks without triggers
Communication	10%	Follows 4.3; concise; precise terminology	Hard to follow; structure deviates

7. Academic integrity

This is an individual assignment. You may discuss general concepts with classmates but the document must be your own work. If you use generative AI tools for brainstorming or editing, you must declare it in an appendix and remain responsible for the content. Submissions are checked for similarity and for consistency with your in-class design exercises.

Appendix A — Interface sketch (non-binding)

The product owner has sketched the following candidate HTTP interface. You may change it.

Operation	Purpose
GET /rooms?campus=&capacity=&from=&to=	Search available rooms (FR-1)
POST /reservations	Create a reservation (FR-2); returns 409 on conflict
PATCH /reservations/{id}	Modify a reservation (FR-2)
DELETE /reservations/{id}	Cancel (FR-2)
POST /block-bookings	Staff block bookings (FR-3)
GET /kiosk/{roomId}/schedule	Kiosk view (FR-4); must support polling or push
GET /reports/utilisation?campus=&month=	Monthly utilisation (FR-6)

Appendix B — Glossary

- Block booking — a staff-created reservation that reserves a room for a recurring period.
- Kiosk — a wall-mounted display outside each study room.

- Utilisation — reserved hours divided by available opening hours.